United International University
Department of Computer Science and Engineering
CSE 4811 / DS 4213: Natural Language Processing

Fall 2026

Practice Problems: Class 02

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1. **Short answers.**
   (a) What is the naive assumption made by the naive Bayes model used for classifying documents?
   (b) What is the difference between discriminative and generative modeling? Explain in one sentence.

   Stanford *CS124 From Languages to Information* **2019** Practice final, Q3 · Solutions

   ETH Zurich *Natural Language Processing* **2021** Final, Q1(e)

2. **Naive Bayes.** Here is a naive Bayes model with the following conditional probability table (each row is that class's unigram language model):

word type	a	b	c
$P(w \mid y=1)$	5/10	3/10	2/10
$P(w \mid y=0)$	2/10	2/10	6/10

$P(y=1)$	$P(y=0)$
8/10	2/10

   Class $y=1$ means the document is about the end of the world, $y=0$ that it is not.
   (a) Consider a document consisting of 2 a's and 1 c. What is the probability that it is about the end of the world? What is the probability that it is not?
   (b) Now suppose that we know the document is about the end of the world ($y=1$). True or false: the naive Bayes model is able to tell us the probability of seeing the document $\vec{w} = (\text{a}, \text{a}, \text{b}, \text{c})$ under the model. If true, what is the probability?
   (c) Consider training and predicting with a naive Bayes classifier for two document classes, without pseudocounts. The word "booyah" appears once for class 1, and never for class 0. When predicting on new data, if the classifier sees "booyah", what is the posterior probability of class 1?
   (d) For naive Bayes with many classes, consider the case where you only care about the ratio between the posterior probabilities for two classes, say C and D. Show that you do not need to calculate the Bayes rule normalizer $p(x)$ to calculate this ratio, $p(y=C \mid x)/p(y=D \mid x)$.

   UMass Amherst *CS485 Natural Language Processing* **2024** Practice midterm, Q3.1, 3.3, 3.4, 8.3

3. **Precision, recall and F1.**
   (a) Suppose you have a binary classification dataset where 20% of the data points have label $+1$ and the remaining 80% have label -1. What are the precision, recall and F1-score of a classifier that always predicts $+1$?
   (b) You evaluate a classifier on a dataset with 200 positive and 200 negative examples. It correctly classifies 160 examples as positive and classifies the remaining 240 examples as negative. What is the recall?
   (c) Provide an annotation of a set of six true items that would give a precision of more than 0.8 and a recall of less than 0.2, and use your answer to explain why the F1-score uses the harmonic mean.

   CMU *10-701 Introduction to Machine Learning* **2024** Practice final, Q4.1 · Solutions

   Stanford *CS124 From Languages to Information* **2019** Practice final, Q5 · Solutions

   Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 2, Q1.2 · Solutions

4. **Accuracy or macro-F1.** A system predicts one label for each of five items. The gold labels and the predictions are:

Gold standard	1	8	4	1	3
System prediction	1	5	4	9	3

What is the most appropriate metric to evaluate the system, if you have the gold label of every item? Explain briefly why, then calculate your metric of choice for the table.

Edinburgh *Accelerated Natural Language Processing* **2025** Tutorial 4, Q2(c) · Solutions

5. **The softmax function.** The softmax function takes an arbitrary vector v as input, with $|v|$ dimensions. It computes an output vector, also of $|v|$ dimensions, whose ith element is given by

$$\text{softmax}(v)_i = \frac{\exp(v_i)}{\sum_{j=1}^{|v|} \exp(v_j)}.$$

   (a) What is the purpose of the softmax function?

   (b) What is the purpose of the expression in the numerator?

   (c) What is the purpose of the expression in the denominator?

   Edinburgh *Foundations of Natural Language Processing* **2026** Tutorial week 4, Q2 · Solutions

6. **Shifting and scaling the scores.**

   (a) A classifier's last layer computes scores $z = (z_1, \ldots, z_n)$, which may be arbitrarily large or small, and the softmax then computes $\hat{y}_i = e^{z_i} / \sum_j e^{z_j}$. For some settings of the scores the basic softmax works poorly. Subtracting the maximum value $\max_i z_i$ from all scores produces more reliable results. Explain why.

   (b) A classifier trained with cross-entropy outputs $\text{softmax}(x)$ from scores $x = (x_1, \ldots, x_d)$. The scores are now modified, either by scaling $x \mapsto \alpha x$ with $\alpha > 0$ or by a shift $x \mapsto x + b\mathbf{1}$ with $b \in \mathbb{R}$, and fed into the softmax as before. How do these two transformations affect the training accuracy?

   MIT *6.036 Introduction to Machine Learning* **2019** Midterm, Q6(c) · Solutions

   EPFL *CS-433 Machine Learning* **2022** Final, Q16 · Solutions

7. **The cross-entropy loss.** You build a classifier that predicts the animal in an image as one of four classes, dog, cat, iguana, mouse, and train it with the cross-entropy loss

$$L_{\text{CE}}(\hat{y}, y) = -\sum_{i=1}^{4} y_i \log \hat{y}_i,$$

   where $\hat{y}$ is the predicted probability distribution over the classes and y is the ground truth vector, zero everywhere except for the correct class (for example $y = (0, 0, 1, 0)$ for iguana).

   (a) Suppose you are given an image of an iguana. If the model correctly predicts the distribution $\hat{y} = (0.25, 0.25, 0.3, 0.2)$, what is the value of the cross-entropy loss?

   (b) After some training, the model now incorrectly predicts mouse for the same image, with the distribution $(0.0,\ 0.0,\ 0.4,\ 0.6)$. What is the new value of the loss?

   (c) The model achieves a lower loss for a misprediction than for a correct prediction. Explain what led to this.

   (d) Given your observation in (c), you decide to train the network with the accuracy as the objective instead of the cross-entropy loss. Is this a good idea? Give one reason.

   Stanford *CS230 Deep Learning* **2018** Midterm, Q3(a) to (d)

8. **Mean loss over a batch.** For a batch of B labeled examples, the mean cross-entropy loss is

$$J_B(\theta) = -\frac{1}{B} \sum_{i=1}^{B} \ln p_\theta(t_i \mid x_i),$$

   where θ denotes the trainable parameters, x_i is input i, t_i is its correct class, and $p_\theta(t_i \mid x_i)$ is the probability assigned to that class. For the original batch, $B = 2$ and these probabilities are $1/4$ and $1/2$.

(a) Compute the mean cross-entropy over the two examples.

(b) Duplicate each example once, making a batch of four. At the same parameter values, does (i) the mean loss change, and does (ii) its gradient with respect to the trainable parameters change? Explain each answer.

Johns Hopkins *601.471 NLP: Self-supervised Models* **2026** Quiz 2, Q2 · Solutions

9. **Why the logarithm.** Why do we train to minimize the negative log likelihood rather than training to minimize the negative likelihood?

   CMU *11-667 Large Language Models* **2023** Midterm, Q2(a)

10. **Counting is maximum likelihood.** Relative frequency estimation is a form of maximum likelihood estimation. Prove that this is true for a categorical random variable. Let X be a categorical random variable that can take values $1, \ldots, k$. Assume we have N independent samples $x_1, \ldots, x_N$ from X, where each x_i takes one of the k values. We consider a family of distributions $\hat{P}_\theta(X = x) = \theta_x$ with parameters θ. The likelihood of the data as a function of θ is

$$L(\theta) = \prod_{i=1}^{N} \hat{P}_\theta(X = x_i) = \prod_{i=1}^{N} \theta_{x_i}.$$

    Show that $P_{\text{MLE}}(X = x) = C(x)/N$, where $C(x)$ is the count of x in the sample $x_1, \ldots, x_N$.

    Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 2, bonus question · Solutions

11. **Generating text from a unigram model.** We are given the following corpus:

```
<s> language is awesome </s>
<s> language is awesome </s>
<s> language and students </s>
<s> awesome students like language </s>
```

    (a) Using a unigram language model without smoothing, estimated from the corpus (count `</s>` as a word, not `<s>`), generate two sentences. To generate a sentence manually, randomly choose a real number between 0 and 1 and use the cumulative probabilities of the words in the vocabulary to select a word; repeat until `</s>` is drawn.

    (b) Can the model generate a sentence that is not in the corpus? Can it generate one with a word that is not in the corpus?

    (c) What decides the length of the generated sentences?

    Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 2, Q2 · Solutions